

PROJECT FORESIGHT

Demand Forecasting & Inventory Intelligence

MODEL EVALUATION REPORT

Current methodology: 5-fold date-based Rolling-Origin Cross-Validation

Evaluation Component	Current Verified Result
52-week Seasonal Naive Baseline WAPE	12.17%
LightGBM Rolling-Origin CV WAPE	8.16%
Reported WAPE Improvement	33.01%
Validation Strategy	5-fold Rolling-Origin CV
Primary Metric	WAPE

This report replaces the previous example-style evaluation that used an 80%/20% train-test split and unrelated example metrics. The current report uses the project's Rolling-Origin CV and WAPE-vs-baseline framework.

1. Executive Summary

Project FORESIGHT evaluates weekly SKU-level demand forecasting using a 52-week Seasonal Naive baseline and machine-learning models. Evaluation is performed using date-based Rolling-Origin Cross-Validation so that validation periods occur after their corresponding training periods.

The current verified results are Baseline WAPE = 12.17% and LightGBM Rolling-Origin CV WAPE = 8.16%. Lower WAPE indicates better forecasting accuracy. LightGBM therefore outperforms the selected baseline.

2. Evaluation Objective

The objective is to determine whether the machine-learning forecasting model provides measurable improvement over a simple seasonal reference forecast. The baseline uses demand from the same SKU 52 weeks earlier.

3. Evaluation Methodology

Component	Current Project Configuration
Forecast level	Weekly SKU-level demand
Target	weekly_units_sold
Baseline	52-week Seasonal Naive
Validation	Date-based Rolling-Origin Cross-Validation
CV folds	5
Primary metric	WAPE
Model compared	LightGBM Regressor

3.1 Rolling-Origin Cross-Validation

Rolling-Origin Cross-Validation expands the historical training window over successive folds and validates only on future dates. This simulates real forecasting and reduces the risk of future-data leakage.

The model-selection rule is based on the lowest Cross-Validation WAPE rather than training-set accuracy.

4. WAPE Metric

Weighted Absolute Percentage Error (WAPE) is the primary evaluation metric for Project FORESIGHT. It measures total absolute forecast error relative to total actual demand.

$$\text{WAPE} = \text{Actual Forecast} / \text{Actual} \times 100$$

Lower WAPE is better. The metric provides an aggregate forecast-error measure relative to demand volume.

5. Current Model Results

Model	CV Method	WAPE	Assessment
52-week Seasonal Naive	5-fold Rolling-Origin CV	12.17%	Baseline
LightGBM	5-fold Rolling-Origin CV	8.16%	Beats baseline

The verified project result is therefore: Baseline WAPE = 12.17%, LightGBM WAPE = 8.16%, with a reported WAPE improvement of 33.01%.

6. Interpretation

LightGBM has a lower WAPE than the 52-week Seasonal Naive baseline. This indicates that the ML model captures additional demand patterns beyond the historical same-week seasonal reference.

For business use, the result supports selecting LightGBM as the production forecasting model when it remains the lowest-CV-WAPE candidate.

7. Why the Previous 80/20 Evaluation Was Removed

Previous Report	Current Report
Training = 80%	5-fold Rolling-Origin CV
Testing = 20%	Future-date validation windows
Example WAPE / R^2 values	Current WAPE vs baseline
Example regression metrics	Project-specific forecasting evaluation

The previous report contained example values such as $R^2 = 0.94$ and different WAPE values. Those values do not represent the current project's selected Rolling-Origin evaluation and are intentionally excluded.

8. Production Model Decision

The production forecasting model should be selected using Cross-Validation WAPE. Under the current verified results, LightGBM achieves 8.16% WAPE versus 12.17% for the Seasonal Naive baseline and therefore beats the baseline.

9. Limitations

- Results depend on the quality and historical coverage of weekly SKU data.
- Accuracy may decrease during unusual promotions, demand shocks, or conditions not represented in historical data.
- Results should be revalidated whenever the dataset or feature pipeline changes

10. Conclusion

Project FORESIGHT uses a consistent forecasting evaluation framework centered on Rolling-Origin Cross-Validation and WAPE comparison against a 52-week Seasonal Naive baseline.

Current verified results: Baseline WAPE = 12.17%, LightGBM WAPE = 8.16%, and reported WAPE improvement = 33.01%. These results provide the appropriate evidence for comparing the production ML model against the project's baseline.

Final evaluation statement: LightGBM outperforms the 52-week Seasonal Naive baseline under the project's 5-fold Rolling-Origin CV evaluation using WAPE as the primary metric.